Applied Agentic AI: Metacognition, Verification, and "In-the-Loop" Learning

Proposed course syllabus · graduate and undergraduate versions

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Course description

The integration of AI into the workplace, classroom, and research labs is taking place – our job as the human-in-the-loop is to ensure that we are optimally utilizing artificial intelligence to the best of our abilities to scaffold, and not replace human verification, innovation, creation, etc.

This course is aimed at employing the best practices for AI-scaffolded endeavors and teaching students how to maximize productivity and mitigate risk when interacting with AI. To do so, we introduce the ideas of **Metacognition**, **Verification**, **Rules, Regulations, and Exemplars**, **Ground Truth**, **Planning Guidance and Oversight**, and **Agentic Performance Amplification**.

Metacognition. In this context, metacognition is the process of thinking about how AI is "thinking". LLMs (large language models) are "stochastic parrots" that, relative to their training, will use what they have in context and memory to answer questions, create websites, etc. to the best of their abilities. Making students aware of how AI "thinks" is fundamentally important to getting the optimal results in AI interactions.

Verification. The future of work will no longer be repeating mundane tasks, but instead in verifying AI results by learning to scrutinize their output. By teaching proper verification techniques, we provide the students with the knowledge to properly use AI to improve output and efficiency.

Rules, Regulations, and Exemplars. The rules, regulations, and exemplars give fast and fluent agents guardrails with which to abide by, and establishing this infrastructure (1) keeps agents in check and (2) promotes consistency and performance.

Ground Truth. A ground truth to AI environments gives students a knowledge base and context with which to use to learn and build – without this, it is impossible to predict where information is sourced from.

Planning Guidance and Oversight. Planning is simple when we write plans for ourselves, but not when we write them for AI execution. We have full context for how we process what we write. AI, on the other hand, does not always understand domain-related knowledge, assumptions, etc. that are pertinent to correctly carrying out a plan. Teaching strategic planning practices with Socratic questioning helps solidify plan structures and provides students with the necessary tools to plan well using AI.

Agentic Performance Amplification. Agentic performance amplification happens when students are able to use agents to build, audit, and critique work, ideas, and models of the students. Utilizing the "council of agents" approach maximizes the differentiation of AI output, even while utilizing one model.

This course was built from my own experience in researching AI, teaching with AI, and building my toolkit through the looped learning approach after performance testing.

Target Audience

- **Graduate version:** master's and early doctoral students. Prerequisite: one course in quantitative methods, statistics or analytics.
- **Undergraduate version:** juniors and seniors. Prerequisite: one course in statistics or quantitative methods.
- No programming course or prior AI experience is required. Students learn to read and test the short scripts the agent drafts.
- Students use any AI assistant or coding agent, such as Claude or Claude Code, ChatGPT or Codex, Google Gemini, or GitHub Copilot. No outcome depends on one product.

Learning outcomes

Students who complete the course will be able to:

1. Understand and be able to explain the processes AI goes through to produce output.
2. Verify results from AI through specific techniques and behaviors taught in the course.
3. Build and iterate on standards agents must meet (rules, regulations, and exemplars) for specific projects.

4. Implement iterative improvement approaches to improve efficiency and performance of AI.
5. Plan agentic swarms for effective workflow, fully understanding what each agent will bring to the table.
6. Interact with real-world problems and use AI as a scaffold to make operational decisions and optimize about a goal.

Modules

1. Metacognition: AI's Brain

- An overview of AI as a stochastic parrot
- The basics of AI context, memory, pointers and retrieval-augmented generation (RAG)
- What an agent reads and what it processes
- Context rot
- Correcting a deliberately flawed analysis
- End-of-session review: what the agent got wrong, and why
- *Verification test:* the missing context behind a wrong answer

2. Verification

- Checking results before they are used or signed
- A verdict on every result the agent returns
- The agent runs the check; a person signs the verdict
- PACE (Parallel Agent Consensus Engine): two agents work the same problem independently, each reviewed by its own coach; a cross-reviewer then compares the two and reports where they agree and where they differ
- *Verification test:* planted errors, the check that finds each, and a verdict

3. Iterative improvement and documentation

- The improvement loop: try, check, record, revise
- Documenting each iteration: what changed, why, and what it fixed
- The correction log as a working record that later sessions read
- Measuring whether a change improved efficiency and performance
- *Verification test:* whether a documented change actually improved the result

4. Rules, regulations and exemplars

- Rules: scripts the work must pass, trusted only after they fail
- Regulations: guidance for judgment, given at every session start
- Exemplars: earlier work to follow
- The learning loop: each correction becomes a rule, regulation, exemplar or exception
- *Verification test:* a rule against a planted case

5. Ground truth

- The source itself on file, not the agent's account of it
- Extracted text checked as real text
- A working link before any citation
- *Verification test:* citations whose source does not say what is claimed

6. Planning guidance and oversight

- The charge, and an inventory of what already exists
- Socratic questioning, one question at a time
- A plan, a critic, and a person's approval before building
- Build, then verify against the plan
- *Verification test:* where built work departs from its plan

7. Agentic performance amplification: the Council of Agents, and checks and balances

- Agents that build, audit and critique work, ideas and models
- The Council of Agents: distinct roles, one question, a chair's report
- Module 4 rules as gates a person must approve
- *Verification test:* what a council's agreement does and does not show

Term project

Each student finds a real-world problem or research topic to address. By providing them the opportunity to delve into a passion project in detail, students are able to gain experience while simultaneously learning about pressure points in the workplace/research when working with applied problems.

Each module adds one piece to the project: a correction log (Module 1), a verdict on every result the agent returns (2), a documented record of each iteration (3), the standards file (4), a source audit (5), the approved plan (6), and a map

of the gates and who may override each one (7).

The project is built on its rules: each time the agent gets something wrong, the student records the fix in the standards file as a rule, a regulation, an exemplar or a one-off exception, and by the end the agent's work must pass every rule, each one shown to fail on a planted case. The project closes with a three-part defense, in which students (1) find the defects planted in a copy of their own work, on paper and without AI; (2) run one of their own rules and show it refusing a planted case; and (3) explain why a gate sits where it does.

How the versions differ

Both versions cover the same seven modules and use the same grading. They differ in three places.

Part of the course	Graduate	Undergraduate
Red team (finding planted defects)	Finds the defects planted in a classmate's project; without a client, also builds a flawed project, with an answer key, for a classmate	Finds the defects planted in a classmate's project
Council of Agents	Designs the agents' roles	Uses roles the instructor supplies, run in class
Source audit (checking citations)	Audits five citations, one of them found by the student	Audits three citations the instructor supplies

Grading

AI is allowed, and expected, on all work outside the room. Half the grade comes from a midterm and a final, taken in class, on paper, without AI. The midterm covers Modules 1 to 4; the final covers all seven. Both versions use the same weights.

Component	Weight	Take-home, AI allowed	In class, AI-free, on paper
Examinations	50%	–	50% (midterm 20%, final 30%)
Module work	15%	5% completion	10% module checks
Term project and its rules	35%	20% graded submission	15% defense
Total	**100%**	**25%**	**75%**

Client data

Each client signs a one-page scope letter that says what the student may see and share. Projects use public, synthetic or client-approved data, and none collects data about people. A client is named only with written consent, never on a public page.